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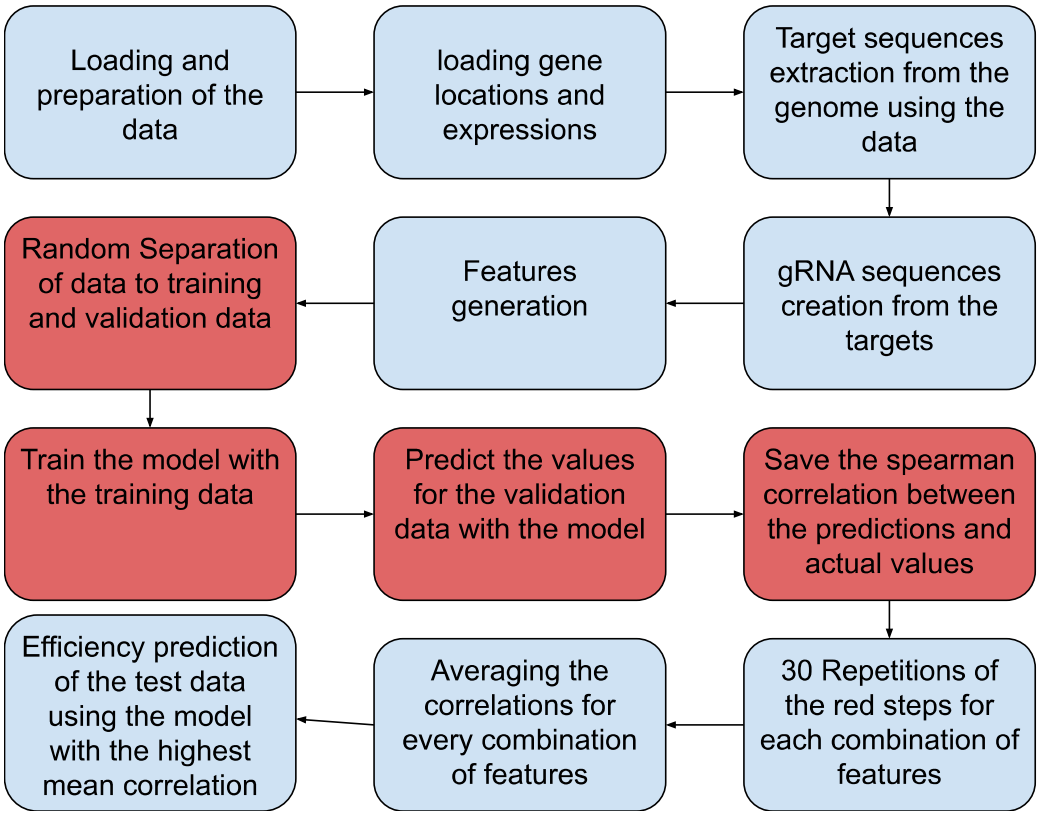
A random forest model using the *fitrensemble()* function in matlab was used for this project due to its ability to handle complex, non-linear relationships between input features and the outcome variable, such as CRISPR efficiency scores or mutation probabilities. By using an ensemble of decision trees, a random forest model can capture interactions and patterns among various features, including gRNA sequences, gene expression levels, and structural properties of target DNA. Additionally, random forests are robust against overfitting and can handle large datasets with high



dimensionality, making them well-suited for analyzing the diverse and multidimensional data typically involved in CRISPR efficiency prediction tasks.

In addition to the mandatory features, we also added a feature measuring GC content of a sequence. DNA sequences with high GC content may exhibit stronger binding affinity to Cas9 compared to sequences with lower GC content. Therefore, sequences with high GC content may have higher Cas9 binding efficacy.

The selection of features were as followed: for every possible combination of features, the model was trained on a random ⅘ of the training data and the spearman correlation

coefficient was calculated between the prediction on a validation set of the remaining ⅕ and their corresponding efficiency values. This process was performed 30 times, and the model with the highest mean spearman correlation coefficient was chosen. The chosen features for our model, which had a mean spearman correlation of 0.26 ± 0.05 were: total amount of each nucleotide and di-nucleotide (AA,AC,AG….), distance in nucleotides to closest gene, expression of the closest gene, RBCS, an nTE. The features were mainly selected to maximize the correlation with the validation set, however it is documented that CUB and nucleotide compositions in general, does affect the cleavage efficacy of Cas9.